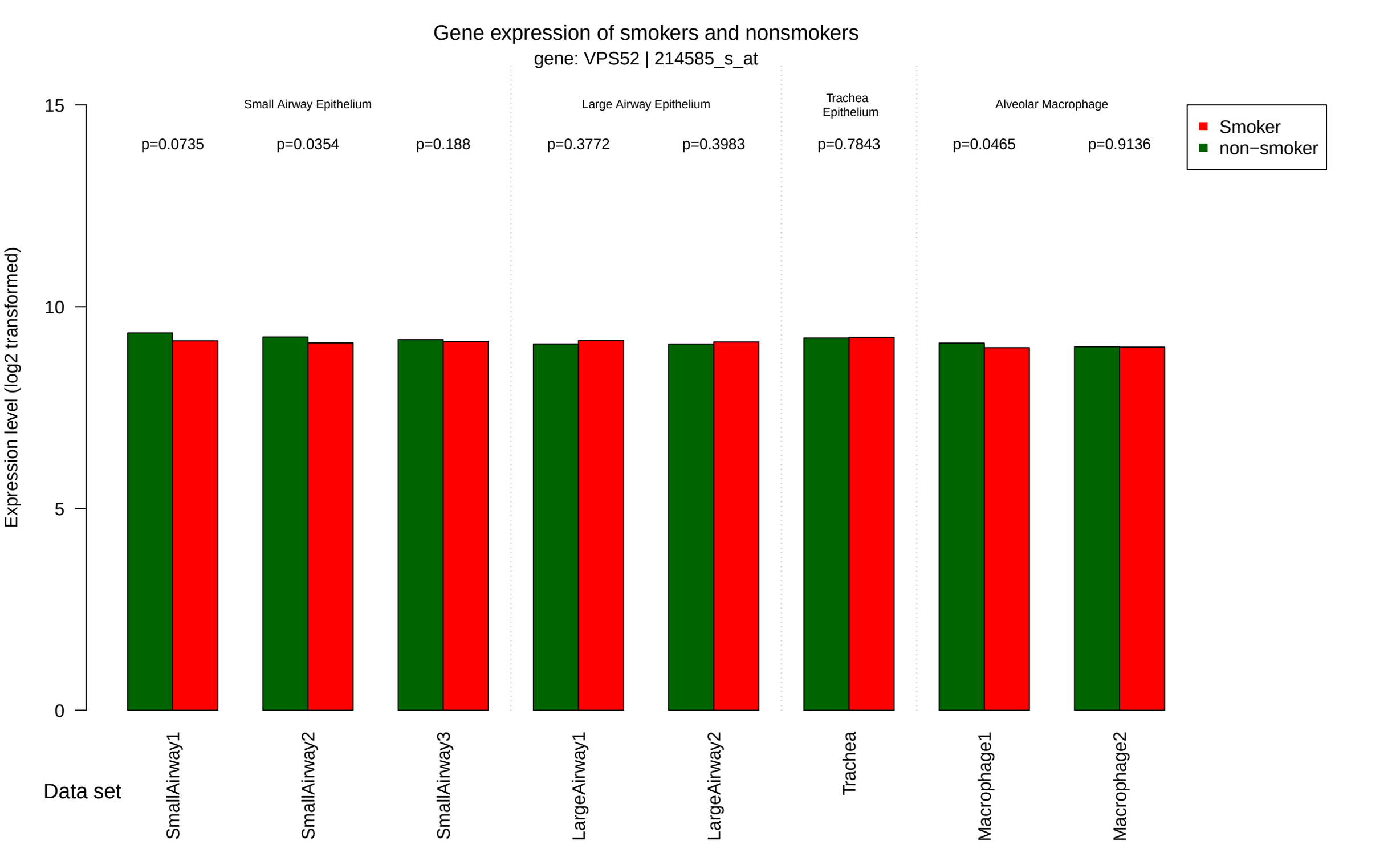
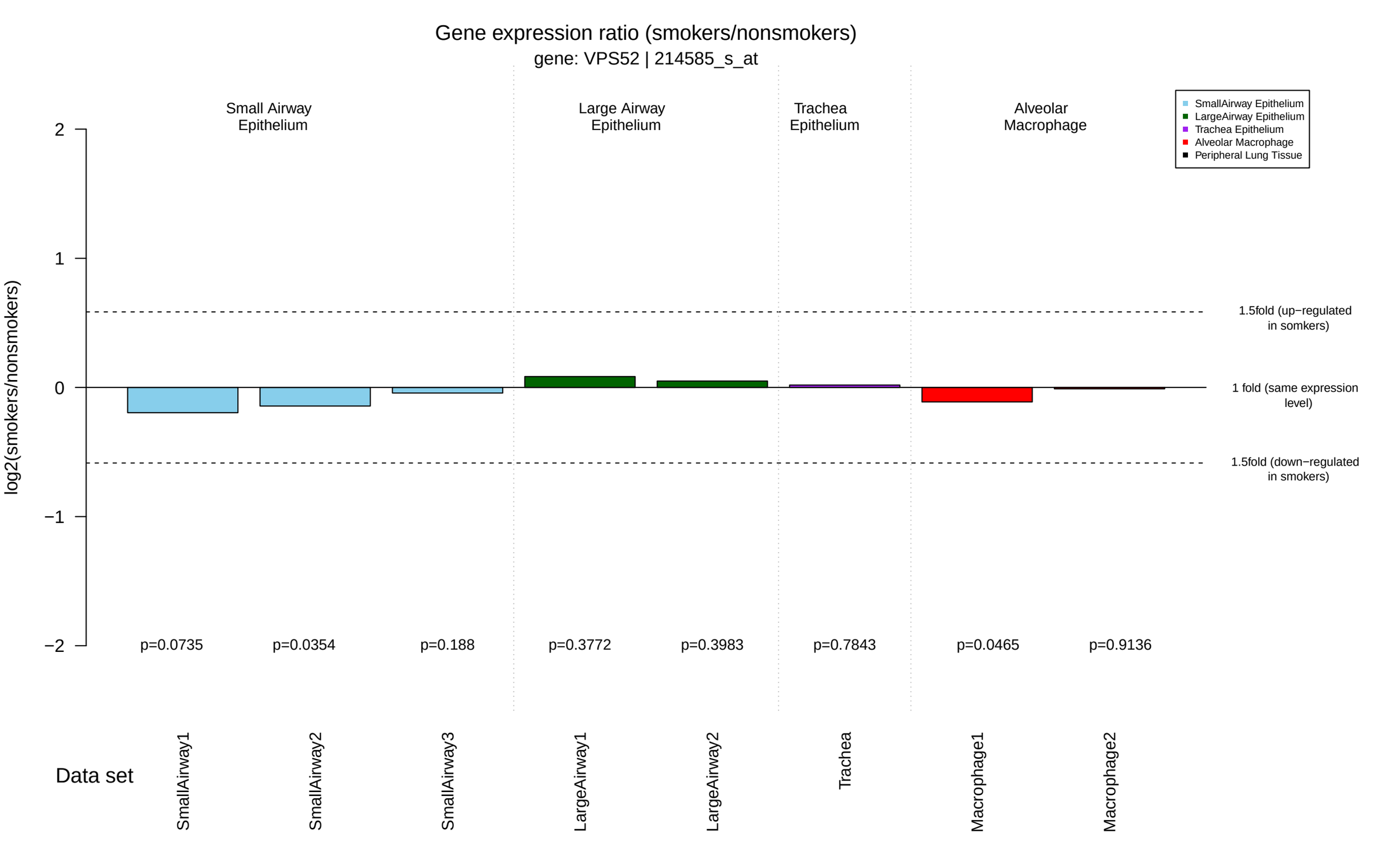
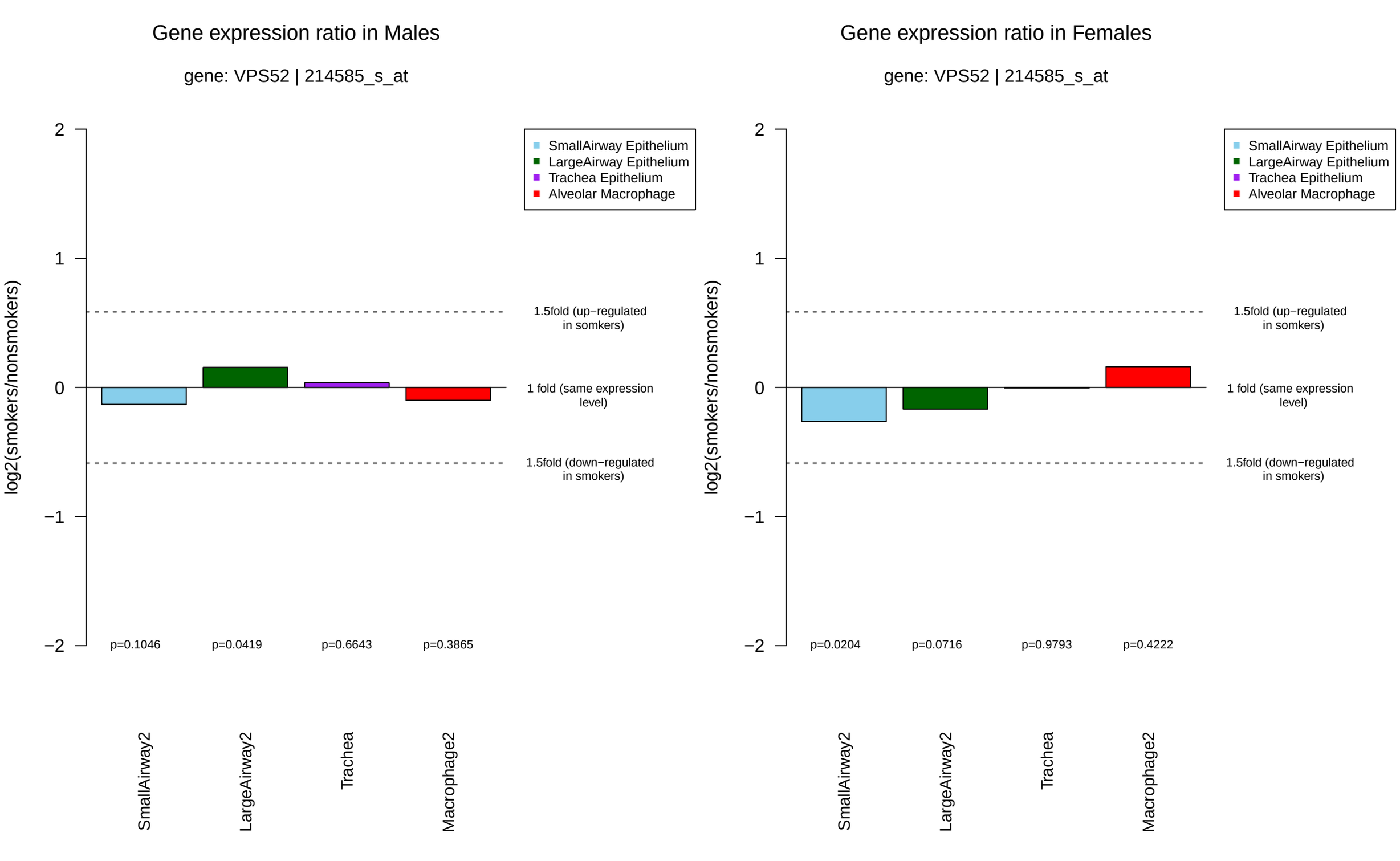
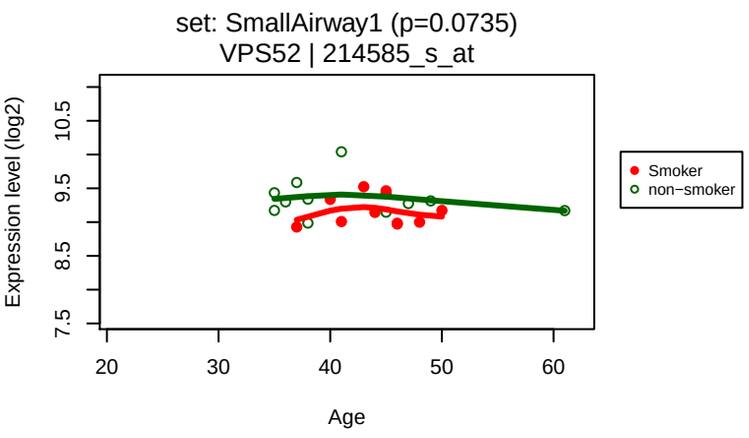
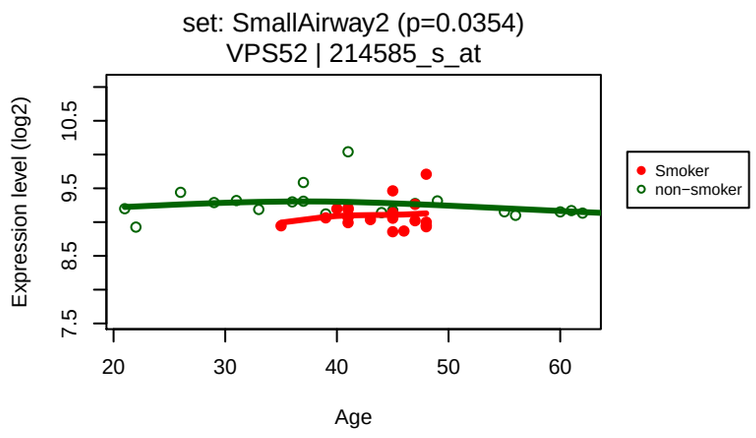
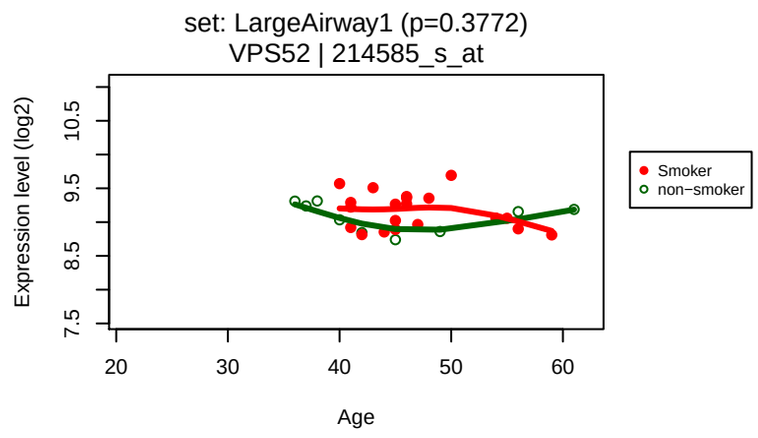

Gene expression of smokers and nonsmokers
gene: VPS52 | 214585_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0735
p=0.0354
p=0.188
p=0.3772
p=0.3983
p=0.7843
p=0.0465
p=0.9136
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

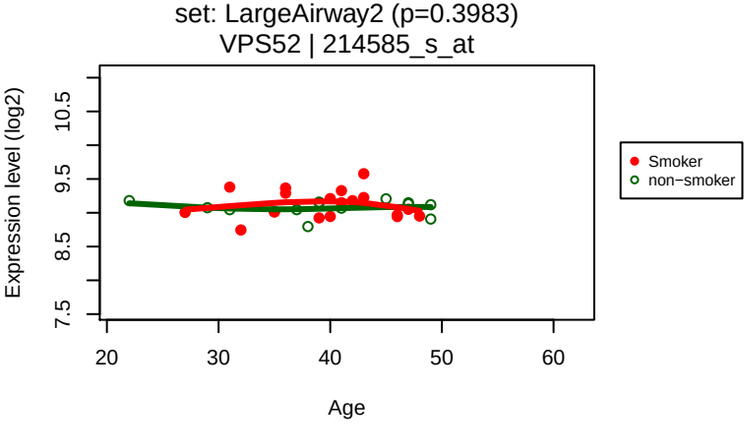

Gene expression ratio (smokers/nonsmokers)
gene: VPS52 | 214585_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0735
p=0.0354
p=0.188
p=0.3772
p=0.3983
p=0.7843
p=0.0465
p=0.9136
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

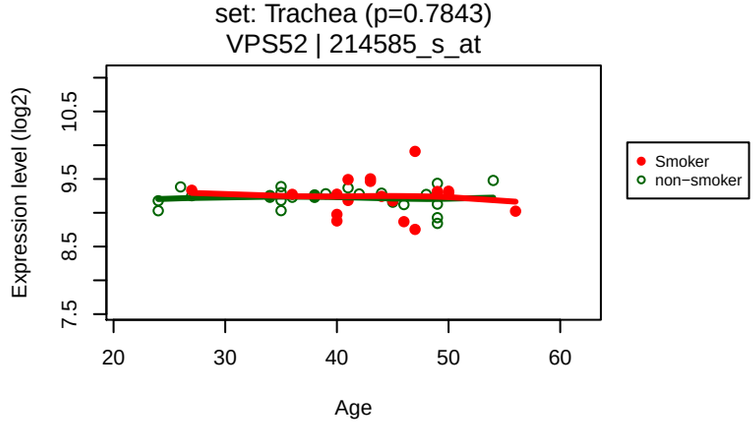

Gene expression ratio in Males
gene: VPS52 | 214585_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1046
p=0.0419
p=0.6643
p=0.3865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VPS52 | 214585_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0204
p=0.0716
p=0.9793
p=0.4222
SmallAirway2
LargeAirway2
Trachea
Macrophage2

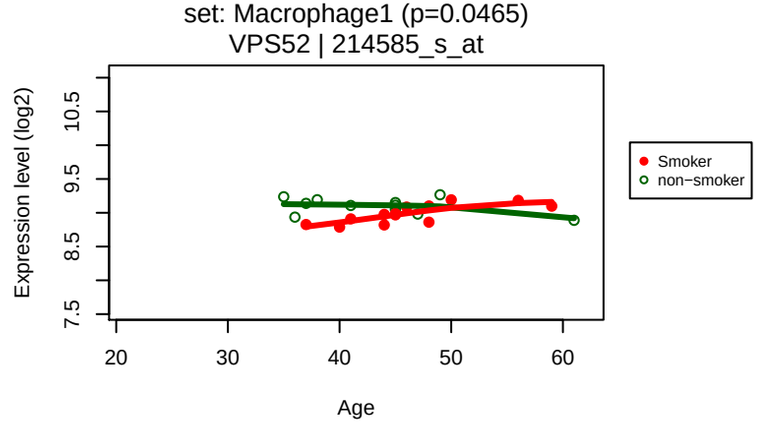

set: SmallAirway1 (p=0.0735)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

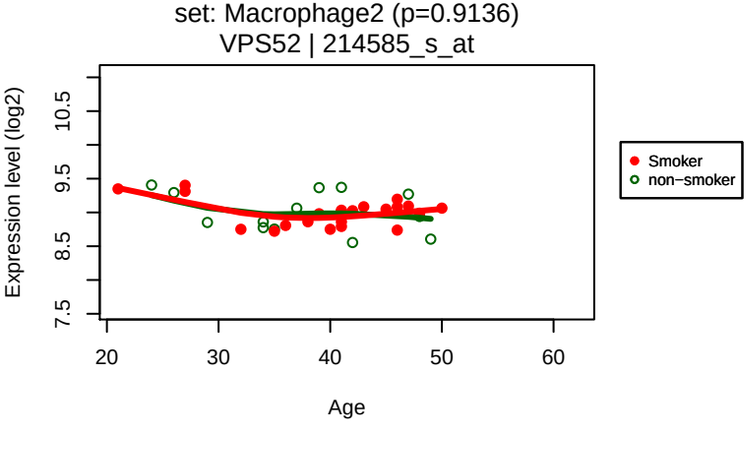

set: SmallAirway2 (p=0.0354)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

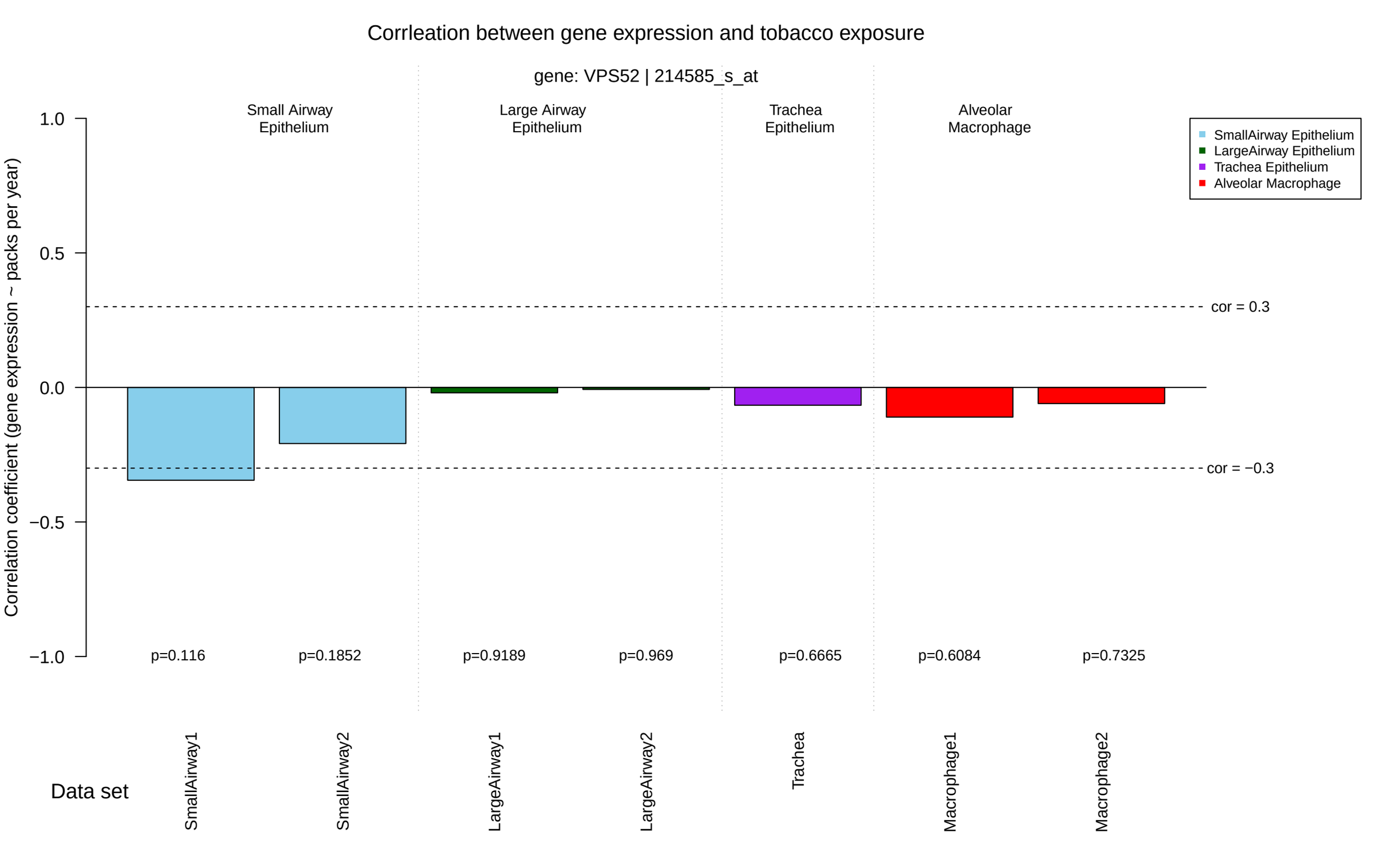

set: LargeAirway1 (p=0.3772)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3983)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7843)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0465)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9136)
VPS52 | 214585_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VPS52 | 214585_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.116
p=0.1852
p=0.9189
p=0.969
p=0.6665
p=0.6084
p=0.7325
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2